

ABSTRACT. We implement the Multi-Secant Limited-Memory BFGS (MS-LBFGS) algorithm proposed by Gubarev in [1].

1. INTRODUCTION

We provide a C++ implementation of the MS-LBFGS algorithm proposed by Gubarev to verify its claimed performance advantages over standard single-secant methods.

2. ALGORITHM IMPLEMENTATION

The algorithm requires an approximation of the inverse Hessian matrix. The limited-memory constraint necessitates a distinct, compact representation for this matrix. Because numerical stability is a priority, we implement the paper's proposed symmetric damping scheme. This involves dynamically dropping older secant pairs when specific overlap conditions are not met. Finally, a numerically stable Armijo line-search optimizer is integrated alongside the quasi-Newton method.

3. BENCHMARKS

A performance comparison with the classic L-BFGS algorithm will be conducted. We will first test the algorithm on random quadratics with a high condition number to observe the multi-secant behavior in an isolated environment. Subsequently, tests using the CUTEst dataset will be performed to evaluate its efficiency in practical, large-scale scenarios.

4. EVALUATION

We will evaluate the overall performance of the algorithm and cross-reference our findings with the results proposed in the original paper. Additionally, profiling of the algorithm's memory usage and computational overhead will be performed.

REFERENCES

1. Gubarev, F.V.: A Multi-Secant Limited-Memory BFGS Method, <https://optimization-online.org/?p=33916>

Email address: gerardo.rocha@cimat.mx

Email address: pedro.llerenas@cimat.mx